

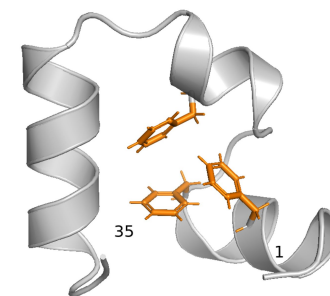
BIOLOGICAL DATASETS FOR COMPUTATIONAL PHYSICS

Markov State Modelling of the HP35 domain

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Instructor: **Prof. Monika Fuxreiter, PhD**

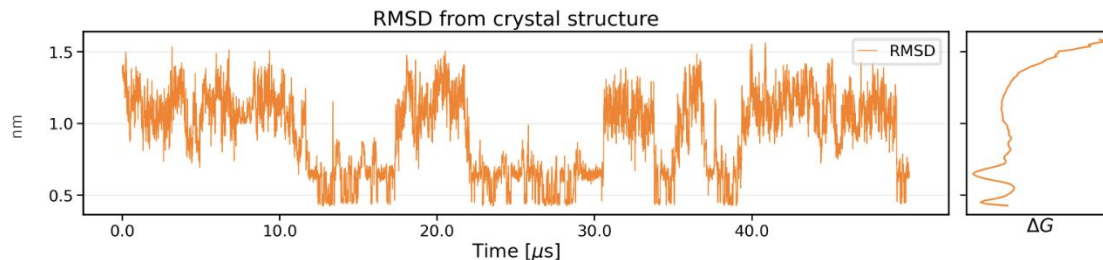
Academic year 2025-2026



PDB 2F4K

J. Kubelka, T.K. Chiu, D.R. Davies et al.
Sub-microsecond protein folding, J. Mol. Biol. 2006

HP35 (Nle/Nle mutant)



Villin headpiece subdomain (HP35)

Established case study

35 residues, ultra fast autonomously folding domain: **< 1 mus** in the Nle/Nle mutant

Biological question:

Does HP35 have resolvable metastable substructure (long-lived states that are structurally well defined) beyond simply folded and unfolded?

Tools: Markov State Modelling (MSM) on a single **300 mus** MD trajectory at T=360K

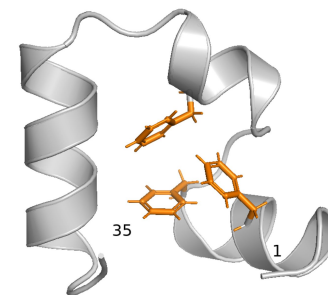
Methodological question: How does the MSM pipeline affect the results?

Data:

original: K. Lindorff-Larsen, S. Piana, R.O. Dror and D.E. Shaw, *How fast-folding proteins fold*, Science 2011

preprocessed: D. Nagel, S. Sartore, and G. Stock, *Selecting Features for Markov Modeling: A Case Study on HP35*, J. of Chem. Theory and Comp. 2023

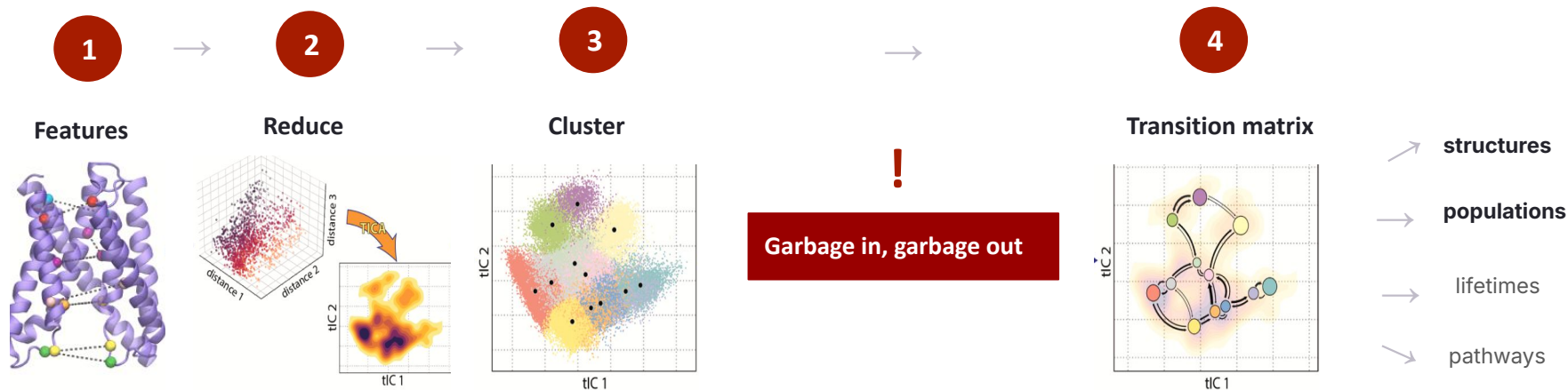
timestep
approx.
200 ps



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The MSM pipeline has no unique recipe (yet)



PCA maximizes variance: high amplitude motion = important

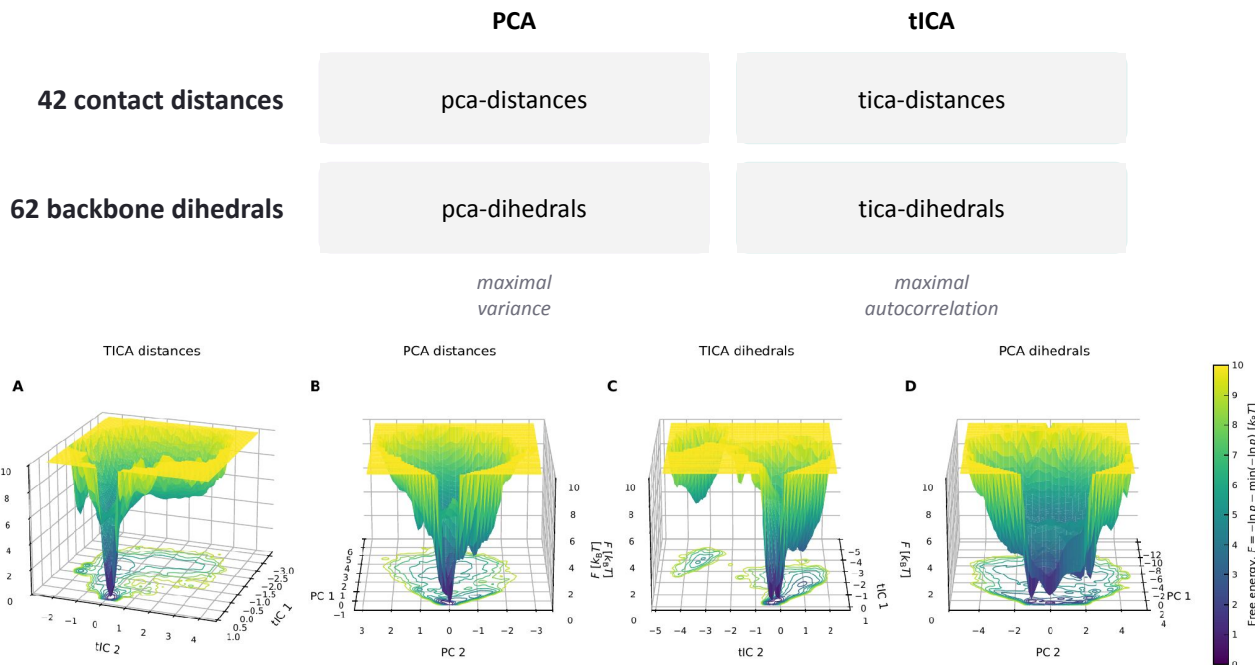
tICA maximizes autocorrelation: slow = important

Clustering algorithms rely on euclidean distance

A good dimensionality reduction maps rapidly interconverting points close in the space: **“kinetic” distance $\approx$ geometric distance**

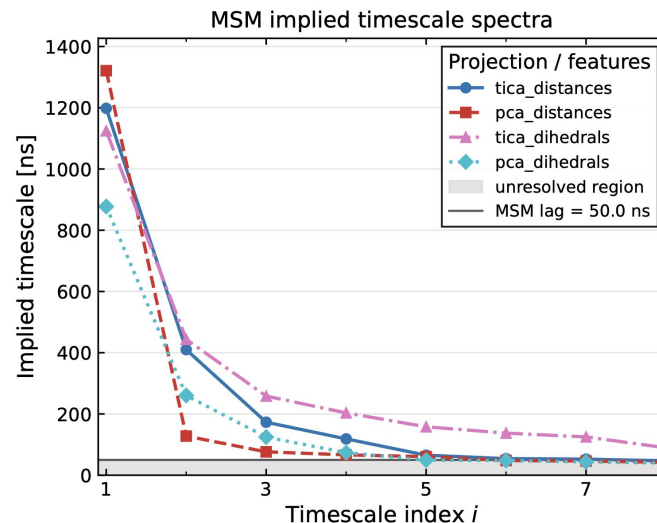
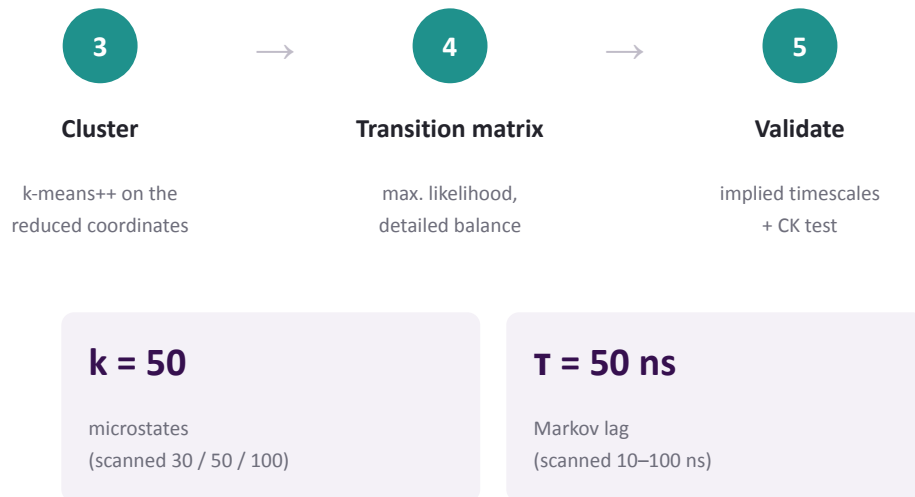
Four sets of reduced coordinates for the transition matrix estimation

Feature choice, dimensionality reduction and clustering are all judgement calls — and different reasonable choices can resolve different metastable states. So instead of committing to one pipeline, I built four.



Free energy on the first two reduced coordinates: one dominant minimum for the distance models, two for the dihedral models, plus a well-separated basin in both tICA projections.

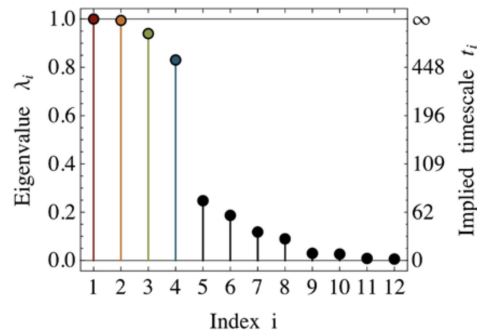
Common parameters across pipelines



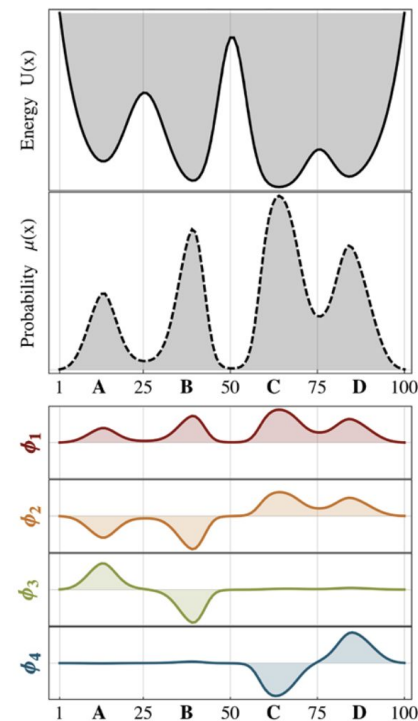
More clusters give resolution but fewer transitions per state and a network that can disconnect; the lag must be long enough for Markovianity yet well below the $0.7 \mu\text{s}$ folding time. All four pipelines share the same k and τ , so whatever differs between the models comes from the features and the reduction — not from the discretisation.

Resulting MSMs were coarse-grained with PCCA++

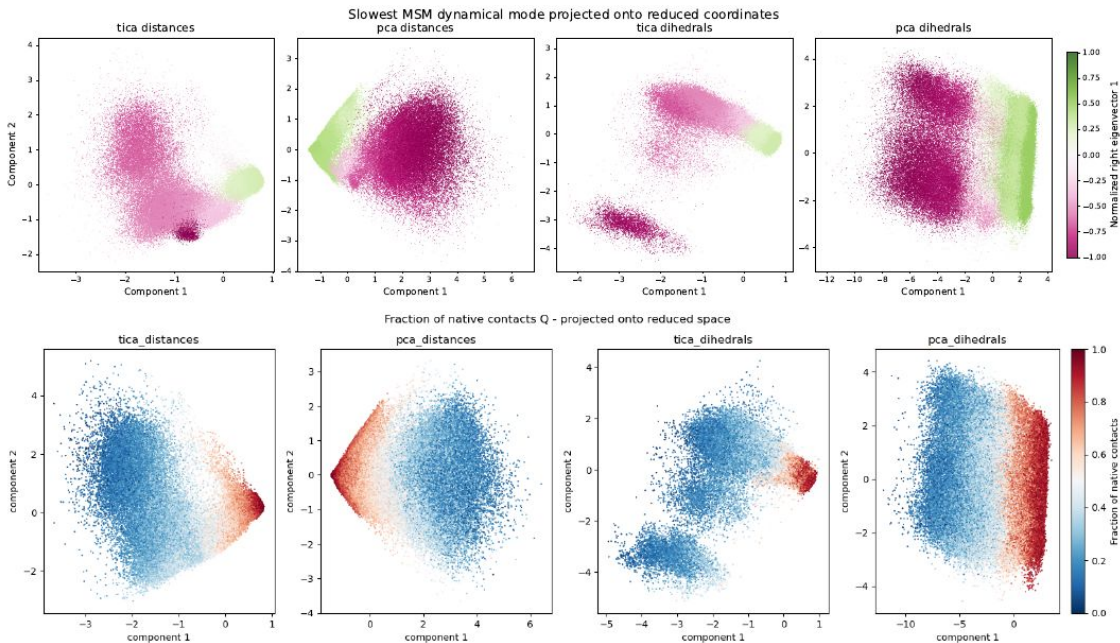
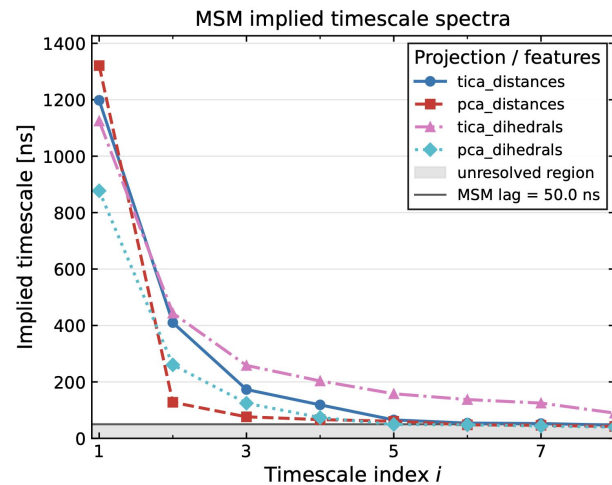
PCCA++ identifies slow processes using the eigenvectors sign patterns



G.R. Bowman, V.S. Pande and F. Noe,
*An Introduction to Markov State Models
 and Their Application to Long
 Timescale Molecular Simulation*
 Springer (2013)



Slowest process separates folded and unfolded conformations



Folded-side basin is distinguished by hydrophobic core and close packing, not by helix formation alone

Consensus folded set frames assigned to S_1 with $\chi > 0.8$ by **all four** models — 61% of the trajectory. Median RMSD 0.65 nm, versus ~1.1 nm for everything else.

Every native contact was then ranked by how sharply it separates the consensus set from the rest — a median difference normalised by the spread of the two distributions.

$$\text{score}_{i,j} = \frac{\text{median}(r_{ij}^{\text{other}}) - \text{median}(r_{ij}^{\text{consensus}})}{[IQR_{ij}^{\text{other}} + IQR_{ij}^{\text{consensus}}] / 2}$$

31–32% unfolded

67–68% folded

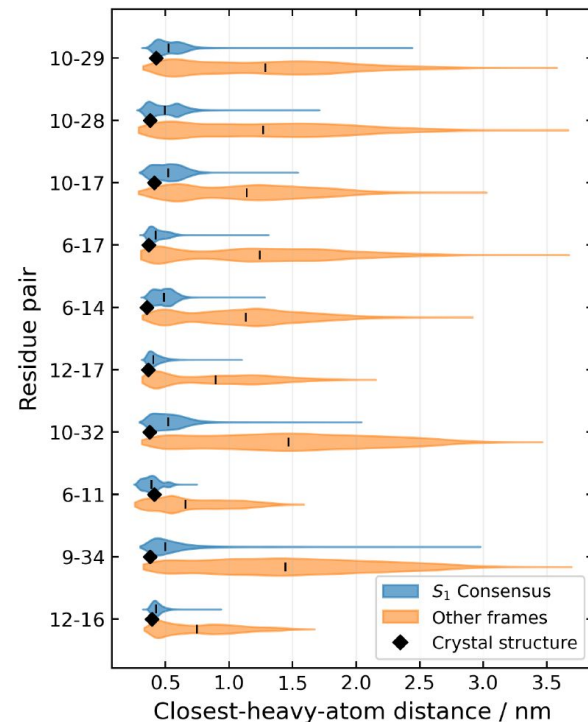
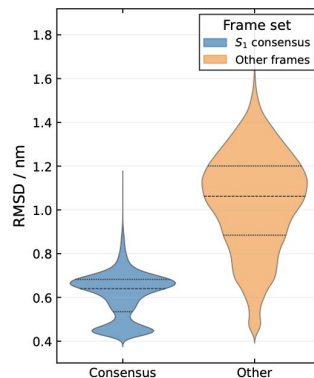
consistent with Piana et al. for this trajectory

Frames assigned to opposite states

1.6 % tica-distances vs pca-distances

2.3 % pca-distances vs pca-dihedrals

≈ 8 % tica-dihedrals vs each of the others



A finer partition: each tICA model finds a different new state

The tICA spectra keep several processes above the resolution limit, so I pushed both tICA models to **three** macrostates.

tica-distances

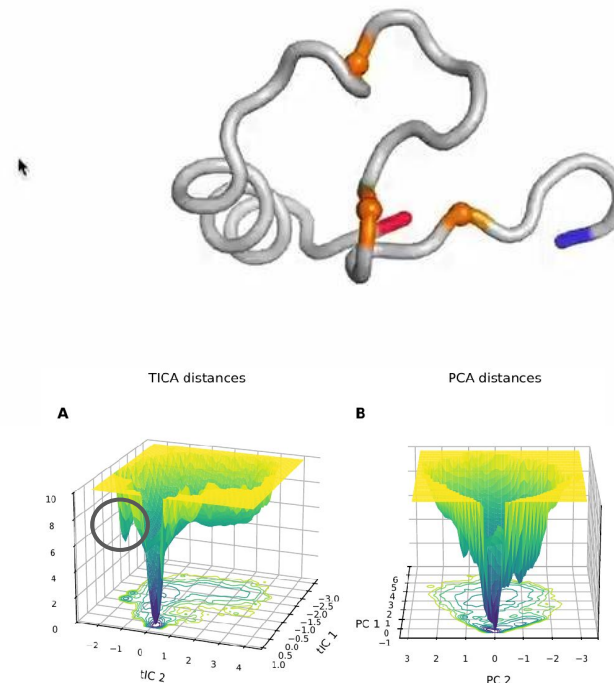
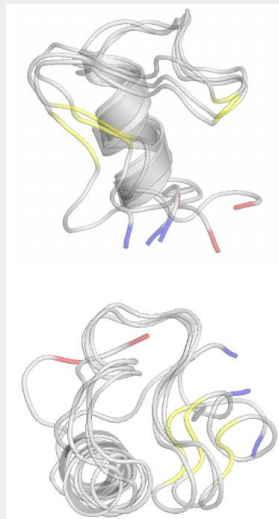
A compact non-native state

S_0 ~1% population · ~1 μ s single visit

Helix 3 stays folded, while residues 6–14 form a stable β -hairpin-like segment **in place of helix 1**.

Distinguished by the backbone dihedrals of Phe6, Lys7 and Gly15 — β region here, α_R elsewhere.

A non-native topology $\Rightarrow$ a plausible off-pathway intermediate, i.e. a putative kinetic trap. Rare but long-lived: one visit in 305 μ



A finer partition: each tICA model finds a different new state

The tICA spectra keep several processes above the resolution limit, so I pushed both tICA models to three macrostates.

tica-dihedrals

A loop rearrangement

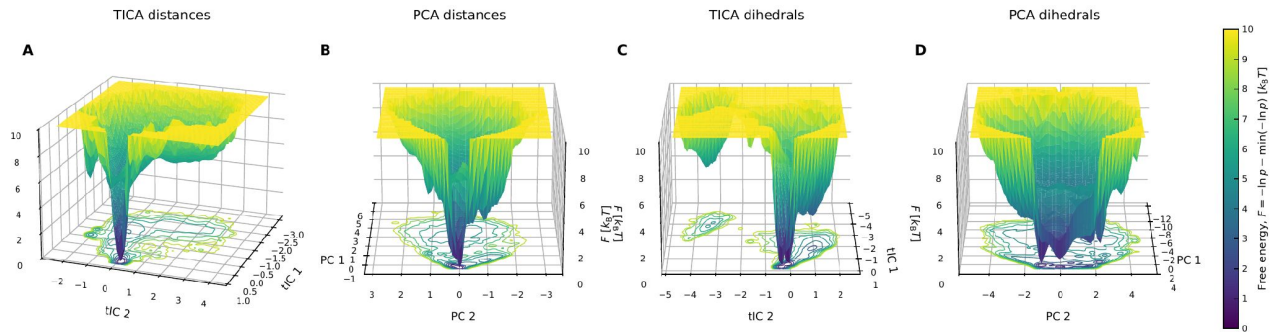
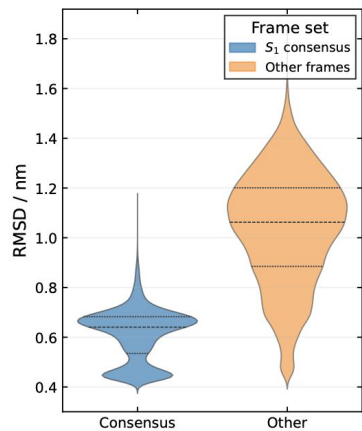
S₀ 3–6% population · 0.32 μ s median residence

The structural signature is much weaker. The only clear difference is the **Leu20–Pro21 loop** joining helices 2 and 3: its ϕ/ψ angles sit in the β region here, and in α_R in the other two macrostates.

Entered only from the unfolded state, and always returns to it — no $S_1 \rightarrow S_0 \rightarrow S_2$ passage is observed.

So this is a local rearrangement inside the unfolded ensemble — not a folding intermediate.

Inside the folded-side basin: a locked and unlocked N terminus



Inside the folded-side basin: a locked and unlocked N terminus

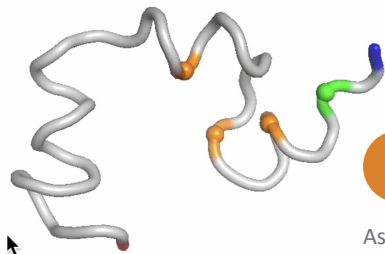
RMSD, the 10–29 contact and the membership distributions of both dihedral models are all bimodal inside the folded basin — yet PCCA++ with three states does not split it. So I subdivided it by hand, at the peaks of the membership probability.



F_1

34–35%

Asp3 in αR · those Leu1 contacts present/absent (fluctuating $\rightarrow$ mobile terminus)

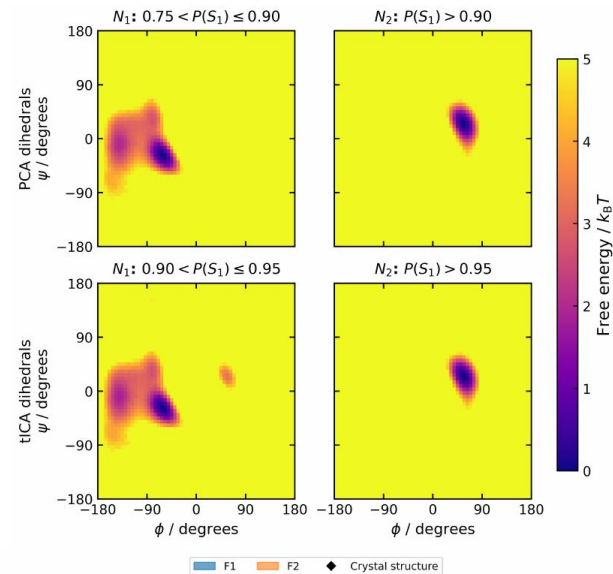


F_2

31–32%

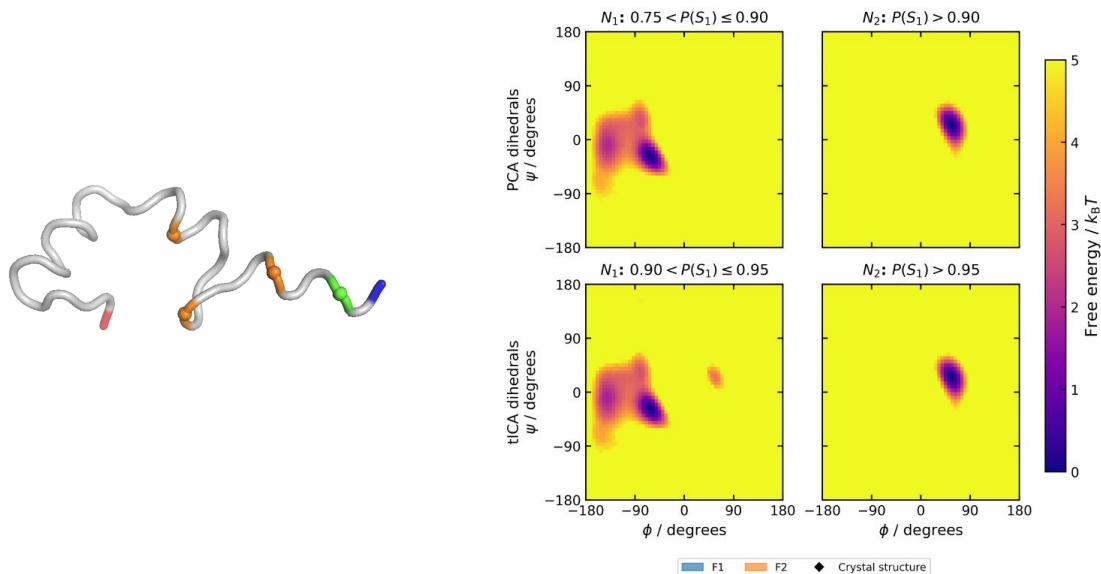
Asp3 in αL
native contacts of Leu1 absent

“unlocked”



(ϕ, ψ) of Asp3 in the two subpopulations, for the PCA (top) and tICA (bottom) dihedral models. The whole distinction rests on one residue.

Inside the folded-side basin: a locked and unlocked N terminus

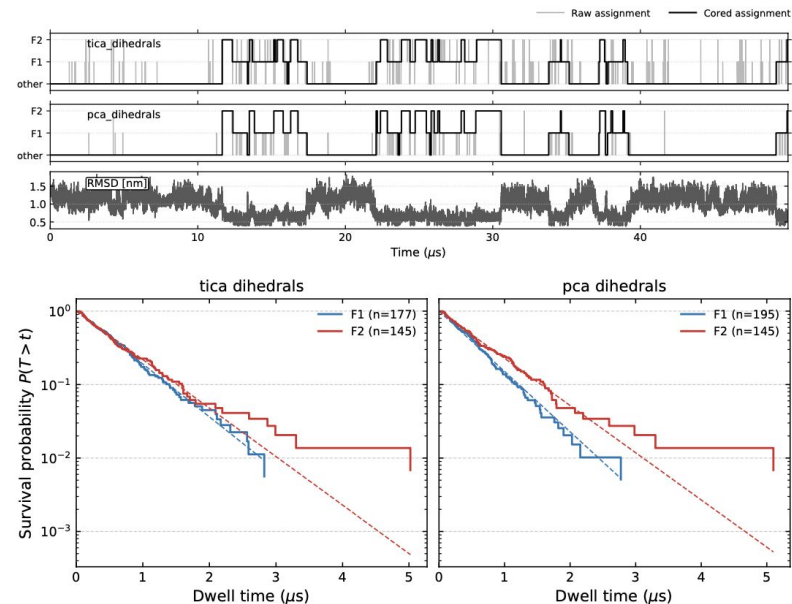


(ϕ , ψ) of Asp3 in the two subpopulations, for the PCA (top) and tICA (bottom) dihedral models. The whole distinction rests on one residue.

Inside the folded-side basin: a locked and unlocked N terminus

Both are metastable in their own right: median residence $\approx 0.4 \mu\text{s}$ (mean $\approx 0.6 \mu\text{s}$), with near single-exponential dwell-time distributions.

Comparable to the loop state that PCCA++ does resolve ($0.32 \mu\text{s}$) — so these are probably genuine kinetic subdivisions that would appear with more macrostates.



Findings about HP35 and model-dependence

1

One dominant process

$\approx 1/3$ unfolded, $\approx 2/3$ folded — separated by the hydrophobic core and helix-1/helix-3 packing, not by secondary-structure content.

2

A $\approx 1\%$ compact non-native state

lifetime $\sim 1 \mu\text{s}$, a β -hairpin-like segment in place of helix 1: a putative kinetic trap, resolved only from contact features.

3

Two substates in the folded basin

of comparable population, differing in the N-terminus — Asp3 in αR vs αL , and the native contacts of Leu1.

The methodological point: one trajectory, four defensible pipelines, and different answers: one says two-state, one finds a trap visible only in contact space, one finds a loop state visible only in dihedral space, dihedral-based models split the “folded” side of the main basin into an almost-folded and a folded conformation, distance-based models do not.

Some insights and lot of room for improvements

1

More macrostates

PCCA++ beyond $n = 3$, especially for the tICA models — the substates found by hand should emerge there.

2

Systematic state comparison

match metastable states across all four pipelines, rather than pairwise at the two-state level.

3

Kinetics, not just populations

transition path theory at the microstate level: dominant folding pathways and fluxes between states.

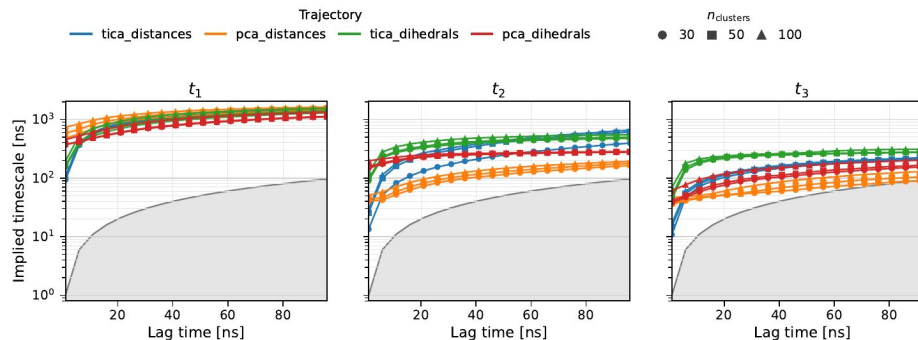
Thank you

Questions welcome — I have backup slides on the discretisation choice, the CK test, coring and the F_1/F_2 dwell times.

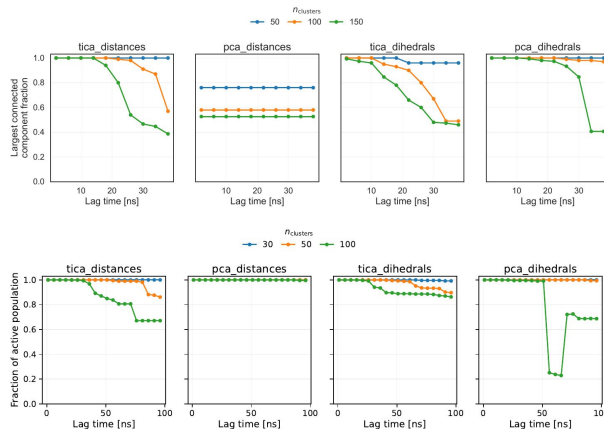
Data github.com/moldyn/HP35-DESRES

Code github.com/miriamzara/MSM

Choice of number of clusters and lagtime



k = 50
 $\tau = 50$ ns

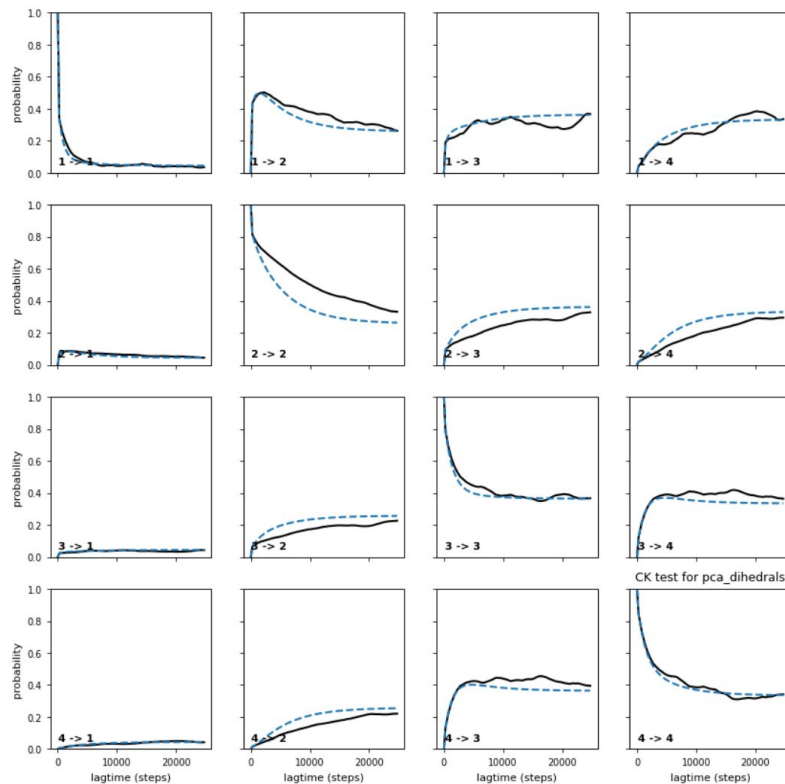


Timescales converge for all four pipelines and depend far more on the pipeline than on k .

Connectivity degrades with more clusters and longer lags: at $k = 100$ the microstate network starts to disconnect.

$k = 50, \tau = 50$ ns is the compromise — and the choice remains partly qualitative.

Chapman–Kolmogorov test

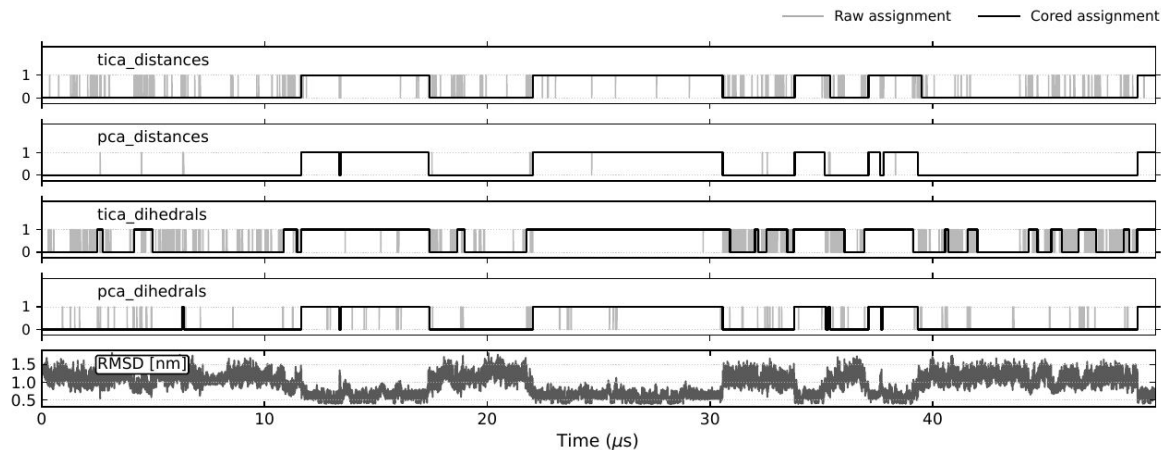


The test. The microstate MSM at $\tau = 50$ ns is lumped into four metastable sets. For a range of lag times $k\tau$, the transition probabilities **predicted** by the $\tau = 50$ ns model (dashed) are compared with those **estimated independently** at each lag (solid).

The two track closely and relax to consistent stationary values, so the model is approximately Markovian at the chosen lag.

Shown for pca-dihedrals; the other three behave similarly.

Dynamical coring removes boundary recrossings



The problem. Assigning every frame to its highest-membership macrostate produces rapid recrossings near a state boundary. These inflate the apparent number of transitions, and mostly reflect boundary noise rather than genuine interstate crossings.

The fix. A transition into a new macrostate is registered only once the trajectory stays there for at least $t_{\min} = \tau = 50$ ns, so only long-lived occupancies survive. Grey = raw assignment, black = cored. All lifetimes and empirical populations quoted here are computed from the cored trajectory.

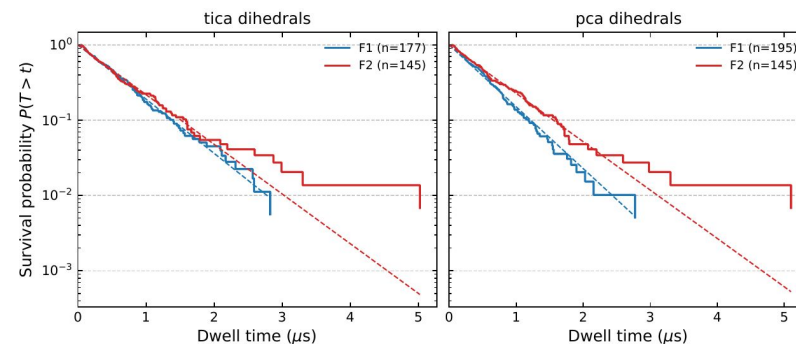
F_1 and F_2 lifetimes

Model	State	N visits	Mean [μ s]	Median [μ s]	Population
tica-dihedrals	F_1	177	0.603	0.463	0.350
tica-dihedrals	F_2	145	0.658	0.435	0.313
pca-dihedrals	F_1	195	0.529	0.412	0.338
pca-dihedrals	F_2	145	0.676	0.480	0.321

Residence times in the two folded sub-basins, from the cored microstate trajectory of each dihedral model.

Yes, in the kinetic sense that matters here. The dwell-time distributions are close to single-exponential (right), which is what a single metastable basin should give, and the median residence of $\approx 0.4 \mu$ s is comparable to the loop macrostate that PCCA++ **does** resolve on tica-dihedrals (0.32μ s).

That PCCA++ resolves one and not the other is not a contradiction: F_1/F_2 are a finer subdivision that should appear with more than three macrostates.



Survival probability $P(T > t)$ with an exponential fit, for both dihedral models.